

SAAM (SEYEDMOHAMMADHOSSEINI) HOSSEINI

CS Ph.D. student, ML systems — checkpointing for multi-tenant LLM training; co-author: USENIX NSDI '26, ACM SIGCOMM '26
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EDUCATION

University of Colorado Boulder

Ph.D. in Computer Science — GPA: 4.0/4.0; advisors: Dr. Mark Zhao & Prof. Sangtae Ha

Boulder, CO

Jan 2025 – May 2029 (expected)

University of Edinburgh

Master in Computer Science (Chevening Scholar) — withdrew to begin Ph.D. at CU Boulder

Edinburgh, UK

2024

Ferdowsi University of Mashhad

B.Sc. in Computer Engineering — GPA: 19.12/20.0, ranked 3rd of 140

Mashhad, Iran

2018 – 2023

RESEARCH EXPERIENCE

Graduate Research Assistant

University of Colorado Boulder — advised by Dr. Mark Zhao & Prof. Sangtae Ha

Boulder, CO

Jan 2025 – Present

- Leading (first author) **Cluster-Wide Checkpointing for Multi-Tenant LLM Training** (in preparation; targeting **EuroSys/ASPLOS 2027**): a controller that sets each job's checkpoint frequency, placement (host memory, local disk, peer nodes' memory, or a remote store), and network bandwidth; validating on an 8×A100 cluster on Jetstream2
- Co-developed **eXpressSFU (USENIX NSDI '26**, 2nd author), a WebRTC SFU that offloads the media data plane — SRTP crypto, packet forwarding, and simulcast/SVC layer selection — onto the Arm cores of an NVIDIA BlueField-2 SmartNIC via DPDK, cutting media-packet latency 8× and compute power up to 60% while serving 3× more concurrent users than the software baseline (300 users at 1080p on a single Arm core)
- Contributed to **DeepSFU (ACM SIGCOMM '26**, 3rd author), real-time deepfake detection as a native SFU service: detects deepfake onsets directly on the encoded VP9 bitstream and offloads crypto and NIC-to-GPU frame movement to a BlueField-3 DPU (DOCA AES-GCM, GPUDirect RDMA, CUDA/NVDEC) — ~144× lower per-frame processing latency and 26× more concurrent conferences at 95.8% precision

PUBLICATIONS

- T. Tran, **S. M. H. Hosseini**, S. Kim, K. Lee, N. Bui, D. Grunwald, S. Ha. “**eXpressSFU: Toward Super-Scalable Video Conferencing with SmartNICs**.” *USENIX NSDI*, 2026.
- T. Tran, S. Ebadi, **S. M. H. Hosseini**, W. Shin, E. Ram, Y. Son, S. Kim, N. Bui, K. Lee, E. Keller, S. Ha. “**DeepSFU: Scalable Deepfake Detection for Video Conferencing**.” *ACM SIGCOMM*, 2026.
- H. Dinh, T. Tran, P. Oli, Y. Son, L. Gutierrez, K. Diaz, **S. M. H. Hosseini**, X. Zhou, S. Ha, N. Bui. “**MAGVIS: A Maglev-based Volumetric Platform for Immersive Telepresence System**.” In preparation, 2026.

INDUSTRY EXPERIENCE

DevOps Engineer

Skysroom — Iran's leading video-conferencing platform

Iran

Mar 2020 – Nov 2022

- Operated production infrastructure through a ~100× COVID-19 scale-up, sustaining platform peaks of ~540K concurrent users across 33K+ simultaneous live sessions (up to 3K participants in a single event)
- Managed production Kubernetes clusters; containerized services with Docker on VMware and bare-metal Linux
- Built GitLab CI/CD pipelines; automated provisioning, configuration management, and deployment with Ansible
- Operated the observability stack (Prometheus/Grafana, ELK/EFK logging) and NGINX/HAProxy load balancing

TECHNICAL SKILLS

Languages: C, C++, CUDA, Python, Java, Bash, SQL, JavaScript

ML systems: PyTorch, NCCL, DeepSpeed, distributed training

Systems & Networking: Linux, WebRTC/SFU (Jitsi Videobridge), raw sockets, GStreamer, OpenMP

Infrastructure: Kubernetes, Docker, Spark, gRPC, GCP, MinIO, GitLab CI, Ansible, VMware, NGINX, HAProxy, Prometheus, Grafana, ELK/EFK

HONORS & AWARDS

Chevening Scholarship, UK Government — one of two recipients selected from Iran (2024)

Dean's & Departmental Excellence Fellowship, University of Colorado Boulder (2025)

Iranian National Computer Olympiad — ranked 33rd nationally (2022)

Iranian National University Entrance Exam (Konkur) — ranked 280th nationally, Mathematics and Physics group (2018)

SERVICE

Reviewer, IEEE Transactions on Mobile Computing (TMC), 2026

Student Volunteer, ACM SIGCOMM 2026

President (elected), CE Student Association, Ferdowsi University of Mashhad (2019 – 2020)